GHRSST Validation Protocol Document (VPD)

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GHRSST Validation Protocol Document (VPD)

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Executive Summary

Validation of GHRSST SST products is essential to provide users with confidence in their content. This document sumamrises the procedures agreed by the GHRSST Internaitonal Science Team to ensure commonality of methods across the GHRSST community.

1 Introduction

A new generation of integrated Sea Surface Temperature (SST) data products are being provided by the Group for High Resolution Sea Surface Temperature (GHRSST). L2 products are provided by a variety of data providers in a common format. L3 and L4 products combine, in near-real time, various SST data products from several different satellite sensors and in situ observations and maintain fine spatial and temporal resolution needed by SST inputs to a variety of ocean and atmosphere applications in the operational and scientific communities. All GHRSST products have a standard format, include uncertainty estimates for each measurement, and are served to the international user community free of charge through a variety of data transport mechanisms and access points that are collectively referred to as the GHRSST Regional/Global Task Sharing (R/GTS) framework.

Major improvements to satellite SST validation are expected to result from increased accuracy and resolution of in situ SSTs from ships and data buoys and independent data from a global, sustained network of ship-borne radiometers to validate the RTM retrievals. Such validation is essential to challenge the retrievals properly and independently. Furthermore, such process studies using accurate in situ radiometry underpin our fundamental understanding of the air-sea interface: ultimately allowing us a reality insight from which we may then expand to a more general framework with a model based approach. Especially in high latitudes, more in-situ measurements are required.

In the past, GHRSST has discussed the possibility of establishing community consensus datasets and data-bases, sampling methodologies and QC procedures concerning the use of in situ SST for satellite calibration and validation. Progress has been limited by the availability of funding. This goal was also identified by the NASA SST ST. Historically, cal/val activities focused on a accuracy validation of an individual satellite sensors shortly after launch by single cruise opportunities. Lessons from the past show that satellite borne sensors may drift, and their stability cannot be taken for granted. To ensure long-term stability of the satellite derived SST products, as e.g., required for climate applications, a long-term cal/val has to be performed.

All GHRSST cal/val activities are coordinated within ST-VAL.

1.1 Scope of this document

This document contains the recomneded GHRSST Validaiton Protocol for assessing the quality of GHRSST products against other SST datasets. The document focus on the what and whay of valdiaiton and not on the how.

1.2 Applicable Documents

GHRSST Development and Implementation Plan, GDIP, 2010-2015

GHRSST User Requirements Document, URD

GHRSST Data Processing Specification GDS v2.0 rev 4

1.3 Reference Documents

C.J. Merchant, J. Mittaz and G.K. Corlett, 2014. Group for High Resolution Sea Surface Temperature (GHRSST) Climate Data Assessment Framework (CDAF), Project Document CDR-TAG_CDAF Version 1.0.4, available from www.ghrsst.org.

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1.4 Relevant URLs

[URL-1] GHRSST Web site http://www.ghrsst.org

2 Acronym and abbreviation list

ATSR	Along Track Scanning Radiometer
ABOM	Australian Bureau of Meteorology
AC	Advisory Council
AHL	Atlantic High Latitude
AMSR	Advanced Microwave Scanning Radiometer
AMSR-E	Advanced Microwave Scanning Radiometer - Earth Observing System
ARC	ATSR Re-analysis for Climate
ARC-Lake	ATSR Reprocessing for Climate: Lake Surface Water Temperature & Ice Cover
ASIP	Air-Sea Interaction Profiler
ATOVS	Advanced TIROS Operational Vertical Sounder
ATSR	Along-Track Scanning Radiometer
AUS-TAG	Applications and User Services Technical Advisory Group
AVHRR	Advanced Very High Resolution Radiometer
BODC	British Oceanographic Data Centre
CAC	Central Assembly Center(s)
CAF	Central Application Facility (EUMETSAT)
CASSTA	Composite Arctic Surface Temperatures
CDOP	Continuous Development and Operations Phase
CDR	Climate Data Record
CEOS	Committee on Earth Observation Satellites
CEOS SIT	CEOS Strategic Implementation Team
CF	Climate Forecast (convention of netCDF)
CMS	Centre de Météorologie Spatiale
CNR	Consiglio Nazionale delle Ricerche (I)
COMS	Communications, Oceanography and Meteorology Satellite
CRED	Coral Reef Ecosystem Division
CSIRO	Commonwealth Scientific and Industrial Research Organisation
CTD	Conductivity, Temperature, Depth (in situ ocean measurements)
DAP	Data Access Protocol
DAS-TAG	Data Assembly and Systems Technical Advisory Group
DBCP	Data Buoy Coordination Panel
DMI	Danish Meteorological Institute
DPF	Data Processing Framework
DUE	Data User Element
DVWG	Diurnal Variability Working Group
EAR	Ecological Acoustic Recorder
EARS	EUMETSAT ATOVS Retransmission Service
EARWiG	Estimation and Retrievals Working Group
EB	Equivalent buoy
ECMWF	European Centre for Medium-range Weather Forecasting
ECV	Essential Climate Variable
EEZ	Exclusive economic Zone
ENVISAT	Environmental Satellite
EO	Earth Observation
EOP	Earth Observation Programme
EOSDIS	NASA Earth Observing System Data and Information System
EPS	EUMETSAT Polar System
ERNESST	European Research Network for Estimation from Space of Surface Temperature
ESA	European Space Agency
ESA CCI	European Space Agency Climate Change Initiative
EUM	EUMETSAT
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites

EUR	European RDAC
fCDR	fundamental Climate Data Record
FNMOC	Fleet Numerical Meteorology and Oceanography Center
FTP	File Transfer Protocol
FY	Feng Yun satellites
GAMSSA	Global Australian High-Resolution SST analysis
GCOM	Global Change Observation Mission
GCOM-W	Global Change Observation Mission – Water GHRSST
GCOS	Global Climate Observing System
GDAC	Global Data Assembly Centre
GDIP	GHRSST development and implementation plan
GDS	GHRSST Data Specification
GDSV	Generic Data Structure Viewable
GEO	Group on Earth Observation
GHRSST	Group for High Resolution Sea Surface Temperature
GMPE	GHRSST Multi Product Ensemble
GODAE	Global Ocean Data Assimilation Experiment
GOES	Geostationary Operational Environmental Satellite
GOMS	Geostationary Operational Meteorological Satellite
GOOS	Global Ocean Observing System
GPM	Global Precipitation Mission
GPO	GHRSST Project Office
G-POD	ESA Grid Processing on Demand
GSICS	Global Space-based Intercalibration System
GTS	Global Telecommunications System
HDF	Hierarchical Data Format
HI	Haiyang satellites
HIRLAM	High Resolution Limited Area Model
HL	High Latitude
HL-TAG	High Latitude Technical Advisory Group
HRDDS	High Resolution Diagnostic Data-set
HRPT	High Resolution Picture Transmissions
HTTP	Hypertext Transfer Protocol
IASI	IR Atmospheric Sounding Interferometer
ICOADS	International Comprehensive Ocean-Atmosphere Data Set
IC-TAG	Inter comparison Technical Advisory Group
IFREMER	Institut français de recherche pour l'exploitation de la mer
IMOS	Integrated Marine Observing System
iQuam	In situ SST Quality monitor
IOC	Intergovernmental Oceanographic Commission
IOOS	Integrated Ocean Observing System
IMPF	Image Processing Facility
IR	Infrared
ISAR	Infrared Sea surface temperature Autonomous Radiometer
IST	Ice Surface Temperature
IWWG	Inland Waters Working Group (was LSWT-WG)
JAXA	Japan Aerospace Exploration Agency
JCOMM	WMO-IOC Joint Technical Commission for Oceanography and Marine Meteorology
JMA	Japan Meteorological Agency
JPL	Jet Propulsion Laboratory
JPL_OurOcean	NASA Jet Propulsion Laboratory
JPSS	Joint Polar Satellite System
KML	Keyhole Markup Language
L2	Level-2 data products

L2P	Level-2 Pre-processed data product
L3	Level 3 data products
L3C	Level 3 collated data product
L3S	Level 3 super-collated product
L3U	Level 3 un-collated data product
L4	Level 4 data product
LAC	Local Area Coverage
LAS	Live Access Server
LEO	Low Earth Orbiter
LML	Low and midlatitude
Ln	Level n of data processing, n=0, 1, 2, 3 or 4
LST	Local Sun Time
LTSRF	Long Term Stewardship and Reanalysis Facility
LWST-WG	Lake Surface Water Temperature Working Group (now IWWG)
M-AERI	Marine-Atmospheric Emitted Radiance Interferometer
MDB	Matchup Data Base
METNO	Norwegian Meteorological Institute
MetOp	Meteorological Operational Satellite
MFC	Monitoring and Forecast Center
MGDSST	Merged satellite and in situ data Global Daily Surface Temperatures
MISST	Multi-Sensor Improved Sea-Surface Temperature
MIZ	marginal ice zone
MMR	Master Metadata Repository
MMS	Multi-sensor match-up system
MODIS	Moderate Resolution Imaging Spectroradiometer
MSG	METEOSAT Second Generation
MTSAT	Multi-functional Transport Satellite
μ-wave	Microwave
MW	MicroWave
MWRI	Microwave Radiation Imager μ-wave
MYO	MyOcean
MyOcean	EU-GMES project for Ocean Monitoring and forecasting
NAR	North Atlantic Region
NASA	National Aeronautics and Space Administration
NASA-SST	The NASA Sea Surface Temperature Science Team
NAVO	US Naval Oceanographic Office
NCDC	NOAA National Climatic Data Center (US)
NEODAAS	NERC Observation Data Acquisition and Analysis Service
NESDIS	National Environmental Satellite, Data and Information Service (US)
netCDF	Network Common Data Format
NGSST-O	New Generation SST for Open Ocean
NIR	Near Infrared
NIRST	New IR Sensor Technology
NIST	National Institute of Standards and Technology
NLSST	Non-linear SST
NMP	New Millennium Program
NOAA	National Oceanic and Atmospheric Administration (US)
NOAA ETL	NOAA Environmental Technology Laboratory
NOC	National Oceanography Centre, Southampton
NODC	NOAA National Oceanographic Data Center (US)
NODC LAS	NODC Live Access Server
NOP	Numerical Ocean Prediction
NOPP	National Oceanographic Partnership Program (US)
NPL	National Physical Laboratory

NPOESS	National Polar-orbiting Operational Environmental Satellite System
NPP	National Polar-orbiting Operational Environmental Satellite System Preparatory Project
NRL	Naval Research Laboratory
NRT	Near-real-time
NSIDC	National Snow and Ice Data Center
NWC SAF	Nowcasting and Very Short Range Forecasting
NWP	Numerical Weather Prediction
OA	Objective Analysis
OceanSITES	A worldwide system of long-term, deepwater reference stations
OCG	JCOMM Observations Programme Area Coordination Group
ODYSSEA	Multi-Sensor Merged High-Resolution SST over Global Ocean at 0.1° Geographical Grid
OE	Optimal estimation
OMP	Office of Marine Prediction (JMA)
OPeNDAP	Open-source Project for a Network Data Access Protocol
OSDPD	NOAA Office of Satellite Data Processing and Distribution
OSE	Observing system experiments
OSISAF	EUMETSAT Ocean and Sea Ice Satellite Applications Facility
OSSE	Observing System Simulation Experiments
OSTIA	Operational Sea Surface Temperature and Sea Ice Analysis
PFV	Pathfinder version
PMEL	Precision Measurement Equipment Laboratory
PNG	Portable Network Graphics
PO.DAAC	Physical Oceanography Distributed Active Archive Centre (US)
POES	Polar Operational Environmental Satellite
PP	Pilot Project
PSBE	Potential Satellite Bias Error
PVP	Product Validation Plot
QA4EO	Quality Assurance for Earth Observation data
QC	Quality Control
QCed	Quality Controlled
QI	Quality Indicator
R&D	Research & Development
R/GTS	Regional/Global Task Sharing framework of GHRSST
R2HA2	Rescue & Reprocessing of Historical AVHRR Archives
RAMSSA	Regional Australian Multi-sensor SST analysis
RAN-TAG	Reanalysis Technical Advisory Group
RD	Reference document
RDAC	Regional Data Assembly Centre
REMSS	Remote Sensing Systems, CA, USA
RFI	Radio Frequency Interference
RMS	Root Mean Square
RSD	Relative Standard Deviation
RSMAS	Rosenstiel School of Marine and Atmospheric Science, University of Miami
RTM	Radiative Transfer Model
RTTOV	Radiative Transfer for TOVS
SAF	Satellite Applications Facility
SEVIRI	Spinning Enhanced Visible and Infrared Imager
SGLI	Second generation GLobal Imager
SI	System International
SLSTR	Sea and Land Surface Temperature Radiometer
SOOP	Ship-of-Opportunity Programme
SOT	JCOMM Ship Observing Team
SOTO	State of the Ocean
SoW	Statement of Work

SPURS	Salinity Processes in the Upper Ocean Regional Study
SQUAM	SST Quality Monitor
SSES	Single Sensor Error Statistics
SSI	Surface Solar Irradiance
SSM/I	Special sensor microwave imager
SSS	Sea Surface Salinity
SST	Sea Surface Temperature
STS	Surface Temperature and Salinity
STVAL	Sea Surface Temperature Validation Working Group
TAC	Thematic Assembly Center
TAG	Technical Advisory Group
TB	Brightness Temperature
TDS	THREDDS Data Server
THREDDS	Thematic Realtime Environmental Distributed Data Services
TIR	Thermal Infrared
TMI	TRMM Microwave Imager
TOVS	TIROS Operational Vertical Sounder
TRMM	Tropical Rainfall Measuring Mission
TWP	Tropical Warm Pool
UKMO	UK Met Office
UNESCO	United Nations Educational, Scientific and Cultural Organization
UNOLS	University-National Oceanographic Laboratory System
UPA	United Kingdom Multi-Mission Processing and Archiving Facility
URD	User Requirements Document
URL	Uniform Resource Locator
US	United States
UTC	Coordinated Universal Time
VC	Virtual constellation
VIIRS	Visible/IR Imager Radiometer Suite
WCRP	World Climate Research Program
WCS	Web Coverage Service
WG	Working Group
WHOI	Woods Hole Oceanographic Institution
WindSat	WindSat is a satellite-based polarimetric microwave radiometer
WMO	World Meteorological Organization
WMS	Web Mapping Service
WPN	White paper on Sea Surface Temperature Error Budget from the NASA Interim Sea Surface Temperature Science Team
WWW	WMO World Weather Watch

3 Overview of GHRSST

The Group for High Resolution Sea Surface Temperature (GHRSST) started in 2002 as one of the pilot projects of the Global Ocean Data Assimilation Experiment (GODAE), and is now the main expert group of users and providers of satellite SST data (Donlon et al., 2007). A new generation of high-resolution (< 10 km) global SST products and services, which have a demonstrated positive impact on ocean and atmospheric forecasting systems, are now provided by GHRSST in near-real-time on a day-to-day basis. Looking forward, GHRSST will continue to improve the quality and provision of high-resolution global SST data, and will also expand its portfolio to include historical satellite SST data to support the generation of SST climate data records for the Global Climate Observing System (GCOS). Further details on all aspects of the GHRSST project can be found at http://www.ghrsst-pp.org.

The group coordinates the harmonisation of a wide variety of SST data streams from satellite and surface based sources that are shared, indexed, processed, quality controlled, analysed and documented within a Regional/Global Task Sharing (R/GTS) framework implemented in an internationally distributed manner. Large volumes of data (currently over 25Gb per day) are harnessed by data services to deliver global high-resolution SST datasets together with meaningful uncertainty estimates for each measurement or analysis grid. Research and development within GHRSST continues to tackle problems such as instrument calibration, algorithm development, diurnal variability, skin temperature deviation, and validation of GHRSST products. Analysis products together with SST anomaly fields are generated each day as part of an ensemble approach to improving analysis systems and providing confidence in analysis outputs. Diagnostic data sets are provided via a web interface to assist in monitoring the quality of the SST retrievals and analysis products at a large selection of critical places in the global ocean.

4 Definition of key terms

The following definitions are used throughout this document:

Error: result of a measurement minus a true value of the measurand. Generally, the “true” value of the error is not known.

Uncertainty: Is a parameter, associated with the result of a measurement that characterises the dispersion of the values that could reasonably be attributed to the measurand (given the measurement, in the light of our understanding of the sources of error in the measurement). Here, the parameter is the standard deviation of the dispersion, which is a confidence of 68% or (k=1).

Discrepancy: The difference between the result and the validation value.

(Relative) Bias: The mean value of the discrepancy.

Accuracy: For the term “accuracy” there seems to be two definitions in common circulation. In RD.150, the Global Climate Observing System (GCOS) considers accuracy to be measured by “the bias or systematic error of the data, i.e., the difference between the short-term average measured value of a variable and the truth” where the average referred to has been sufficient to render the random uncertainty in the measured value negligible. In contrast, the definition from the Guide to Uncertainty of Measurement (GUM; RD.191) is also used, whereby accuracy is “the closeness of agreement between the result of a measurement and a true value of a measurand” and therefore a measurement can be inaccurate either by virtue of a large systematic error or because it has a large random uncertainty. We find it useful to have a term available that distinguishes systematic and random uncertainty, and therefore in SST_CCI documents accuracy refers to the estimated magnitude of the systematic error (true bias).

Precision: The difference between one result and the mean of several results obtained by the same method, i.e. reproducibility (includes non-systematic errors only).

Calibration: The process of quantitatively defining the system response to known, controlled system inputs

Validation: The process of assessing by independent means the quality of the data products (the results) derived from the system outputs.

Skin Sea Surface Temperature (SST_{skin}): The temperature measured by an infrared radiometer typically operating at wavelengths 3.7-12 μm (chosen for consistency with the majority of infrared satellite measurements) that represents the temperature within the conductive diffusion-dominated sub-layer at a depth of ~10-20 μm.

Sub-Skin Sea Surface Temperature ($SST_{subskin}$): The subskin temperature represents the temperature at the base of the conductive laminar sub-layer of the ocean surface.

Depth Sea Surface Temperature (SST-depth): Measurements of water temperature beneath the $SST_{subskin}$, measured using a wide variety of platforms and sensors such as drifting buoys, vertical profiling floats, or deep thermistor chains at depths ranging from 10^{-2} - 10^{3} m. Here, the depth will usually be that associated with a drifting buoy (of order 20 cm) or a moored buoy (of order 1 m).

5 GHRSST SST Products

GHRSST provides two major types of global SST products, Level 2 pre-processed (L2P) and Level 4 analysis (L4). The L2P product allows the many different satellite SST data products to be made available to users in a common format, and includes ancillary information to assist a user in assessing whether or not the data are “fit for purpose”. For every input data stream, GHRSST produces an output L2P product that augments the existing SST and geographic information with an estimate of the measurement bias and uncertainty, surface wind speed, aerosol optical depth, surface solar irradiance, sea ice concentration, the time of observation and a set of quality control flags. The L2P data format is NetCDF, following the Climate Forecast (CF) v1.0 convention. The L4 analysis products provide merged, gridded and gap-free SST datasets, which are produced from an optimal analysis of all available satellite and surface-based SST data within the preceding 24 hour period. The L4 data specification is also CF-compliant NetCDF.

5.1 Uncertainty information

A key feature of the GHRSST L2P data product is the provision of Single Sensor Error Statistics (SSES). The SSES provide a bias and standard deviation in comparison to a reference dataset, as well as three quality indicators that together allow the user to make an assessment as to the suitability of a particular satellite SST estimate for their specific application. In an ideal case, the derivation of satellite SST uncertainties would require:

1) A complete traceable characterisation to agreed national standards of the satellite instrument and SST retrieval algorithm, at all times throughout the lifetime of the mission, and

2) A suite of global traceable reference data points that preserve the nature of the satellite SST, are of an accuracy and precision that is higher than the satellite sensor, and are provided throughout the mission.

In addition, if the sensor is part of a series, there should be a sufficient overlap period between successive sensors to allow for a robust characterisation of the inter-sensor period, using the same traceable reference data points for both sensors.

The bias and standard deviation calculated from the comparison to the reference dataset are derived from a database of match-up coincidences produced within predefined spatial and temporal limits; the current GHRSST match-up limits are ±25 km, and ±6 hours. The bias and standard deviation calculated from such a comparison do not provide the uncertainty of each dataset individually, but are simply the mean bias and combined uncertainty of a two dataset comparison. Consequently, the resulting statistics are often dominated by real changes in the SST that can occur within the predefined spatial and temporal limits. Recently, a new method of multi-sensor match-up processing has been proposed that aims to deduce the uncertainty of an individual dataset, providing it is bias free (O’Carroll et al., 2008).

6 SST Validation

6.1 Validation uncertainty budget

The traditional approach to satellite uncertainty estimation involves the generation of a match-up dataset (MD) of coincidences between the satellite and a reference dataset produced within predefined spatial and temporal limits; for example, the current GHRSST-recommended match-up limits are ±25 km, and ±6 hours, although some investigators choose to use other limits. There are few assessments of the spatial and temporal limits for generating match-ups in the literature. Minnett determined that limits of 10 km and 2 hours would introduce an error of up to 0.2 K but this was for a very specific area of the Atlantic Ocean with relatively high temperature variability. Chapter 5.2 defines more stringent criteria for ship-borne radiometer measurements based on the work of Wimmer et al.. In general, the spatial limit should be minimised by using the smallest possible spatial window. Temporal differences, no matter what limit is set, should be minimised using a diurnal variability model.

Following generation of the MD, a statistical analysis is then carried out on the MD, from which the difference (the reference data are often erroneously treated as if they are truth, i.e. they are free of error) and standard deviation are usually calculated. Of course the difference and standard deviation calculated from such a comparison do not provide the uncertainty of each dataset individually, but are simply the mean difference and combined uncertainty of a two-dataset comparison. Consequently, the resulting statistics are often dominated by real changes in the SST that can occur within the predefined spatial and temporal limits (geophysical effects) and these can be significant as determined by Minnett. An alternative method of multi-sensor match-up processing was demonstrated by O'Carroll et al., which aimed to deduce the uncertainty of an individual dataset from a 3-way analysis, assuming each dataset to be bias free. However, this approach does not properly account for geophysical differences between the satellite and reference datasets and also assumes zero correlation between datasets.

When considering potential reference sources consideration must be given to the depth of the SST being assessed. For satellite SST retrievals produced from infrared radiances, the SST is equivalent to the temperature at a depth of ~10 µm and is referred to as SST_{skin}. The deviation between the skin and the sub-skin reduces to a mean bias of -0.17 K when the surface wind speed is > ~6 ms^{-1}, and so surface wind speed observations also form an essential component of any reference data set for satellite SST uncertainty determination. Ideally, the reference source for assessing the quality of the satellite data should be a measurement at a depth that is as close as possible to that provided by the satellite. Indeed, where possible, it should be the same as that provided by the satellite, which is currently only possible for infrared sensors using ship-borne IR radiometers. The match-up problem is made more complex by the fact that the water surface is a dynamic system and the nature of the sea-surface is such that it is hard to ascertain a meaningful mean condition of the sea-surface from a point in situ measurement over an entire satellite pixel.

Taking all of these factors into account is essential when assessing a SST CDR. We therefore define a validation uncertainty budget (i.e. the standard deviation of a Type A statistical assessment of comparisons between satellite and reference data, σ_{Total}) as:

$$\sigma_{Total} = \sqrt{\sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \sigma_4^2 + \sigma_5^2} \quad \cdot \quad (1)$$

where the terms will be defined and discussed below. This approach is similar to that of Wimmer et al. [28] who defined an error model for comparisons between ship-borne radiometers and satellite SSTs. Here, we adapt the approach of Wimmer et al. to cover all potential reference sources and represent the contributing terms as uncertainties rather than absolute errors.

At this point it is useful to briefly consider the difference between error and uncertainty. The Guide to the Expression of Uncertainty in Measurement (GUM) defines: **measurand** as a particular quantity subject to measurement. In this chapter, the measurand is either TOA BT (Level 1b data) or SST (for Level 2 and higher data). Very few instruments directly provide a measure of the measurand, and usually detect a quantity from which the measurand is derived. As an example, in this chapter, we focus on instruments that detect infrared light and it is from these measurements we estimate the temperature of an object.

Any procedure to determine the value of a measurand will be inexact to some degree due to uncertainty introduced into the procedure at some point. The difference between a measurement and the value of the measurand is called the **error**. In remote sensing, the word 'error' has often been used to describe a numerical value that estimates the variability of the error if a measurement is repeated (i.e. a width of the distribution of possible errors) and this leads to confusion as to what the quoted error actually means. To avoid any misunderstanding, the GUM separates these definitions and defines:

- **error (of measurement)**: result of a measurement minus a true value of the measurand, and

- **uncertainty (of measurement)**: is a parameter, associated with the result of a measurement that characterizes the dispersion of the values that could reasonably be attributed to the measurand.

The “true” value of the measurand is often not known but can be estimated using statistical methods (which the GUM refers to as a Type A assessment) or empirical methods (which the GUM refers to as a Type B assessment) and can be considered as having one or more random components and/or one or more systematic components. The effects causing each component of error each give rise to a component of the overall uncertainty, so that the total uncertainty arises from a mix of random and systematic effects.

For satellite-retrieved SST, this neat division of error/uncertainty into random and systematic components is too simplistic. In reality, there is a spectrum of sources of error with greater or lesser degrees of spatio-temporal correlation between measurements. These correlations matter if averages or trends of satellite measurements are to be formed with appropriate attached uncertainty estimates. Also, effects can be systematic when viewed at a particular scale, while at a larger scale they can actually be random.

We now look at the terms in equation (1) in a bit more detail. The main components are:

- Satellite uncertainty (σ_1): This is the uncertainty of the satellite measurement and should be provided with the product. It will usually vary pixel by pixel and will depend on a number of parameters affecting the retrieval (see Chapter 4.3).

- Reference uncertainty (σ_2): This is the uncertainty of the reference measurement and is generally unknown for individual measurements but can be estimated at the dataset level (see next section).

- Geophysical uncertainty: spatial – surface (σ_3): This is the uncertainty arising from comparing a point measurement to a satellite footprint. For any single match-up it is systematic but can be considered to be pseudo-random for a large enough dataset where the full variability of the ocean has been sampled. This term can also be reduced through averaging of the satellite data (e.g. sample 5 by 5 instead of 1 by 1) as suggested by Embury et al., and by necessity includes any uncertainty in geolocation of both the satellite and the reference measurement (which for drifters can be up to 8 km when, as is common, location reports are available to only 1 decimal place).

- Geophysical uncertainty: spatial – depth (σ_4): This is the uncertainty arising from comparing SSTs at different depths. Again, it is systematic for a single match-up and can be reduced by using a model of the upper ocean temperature as demonstrated by Embury et al. Likewise, it can be considered pseudo-random for a large enough dataset whereby the full variability of the atmosphere and ocean has been sampled.

- Geophysical uncertainty: temporal (σ_5): This is the uncertainty arising from comparing SSTs at different times of day (it is not always possible to get exact temporal matches between satellite and reference). This will be systematic for single match-up and may be pseudo-random for a large dataset if the time differences are reduced with a model as demonstrated by Embury et al. The size of the time window is quite important as if it is too large then one differentially samples both heating and cooling of the ocean across its diurnal cycle (see for example [11]), which will then result in a bias in the calculated statistics.

A further consideration is the actual statistical analysis itself. In a Type A statistical analysis you are fitting a model to the available data. Usually, the model that is fitted is a standard Gaussian Probability Density Function (PDF) from which the mean and standard deviation are then calculated. Unfortunately, this approach has no resilience to outliers, i.e. data points that do not conform to the normal distribution. Outliers arise due to any spatial and temporal miss-match errors, and also issues with satellite and reference data quality. In particular, issues with imperfect cloud screening can lead to large uncertainties in the retrieved SST. Consequently, we expect to find a certain number of outliers in our analysis that we must a) ensure do not overtly distort the final uncertainties, and b) estimate in magnitude and, if possible, eliminate using clearly defined criteria.

There are several approaches to evaluate the impact of outliers on the calculated statistics. An accepted way for single sensor assessment is to first reject all match-ups outside of 3-sigma from the mean (3 * the calculated standard deviation, SD) before recalculating the statistics. A 3-sigma filter is somewhat arbitrary but can help to estimate the level of outliers in the data, and has been used widely in AATSR validation. A critical drawback with this approach is that if there are no outliers in the actual data, then a 3-sigma filter would simply throw away good data and so we will not consider the 3-sigma filter any further.

An alternate way to dealing with outliers is to adopt the approach of Merchant and Harris and use robust statistics. In robust statistics the influence of outliers is reduced by using a median estimator (M-est) and median absolute deviation (MAD) of the distribution, the latter being scaled to the equivalent of the normal SD and referred to as the robust standard deviation (RSD). Finally, there is the approach of fitting a Gaussian probability distribution function to the histogram of the differences between satellite and corresponding reference dataset. This latter approach is most similar to fitting a model for a Type A uncertainty estimation.

Each of these methods can be investigated further by calculating them individually for a set of match-up data, and comparing the results to the normal standard deviation. In Figure 6-1 we show a typical distribution of match-up differences between the NOAA-19 AVHRR and drifting buoys (in blue) in waters around Australia. Over plotted on Figure 6-1 are Gaussian PDFs from normal statistics (in green), robust statistics (in orange) and a linear least square fit of a Gaussian PDF to the data (in red).

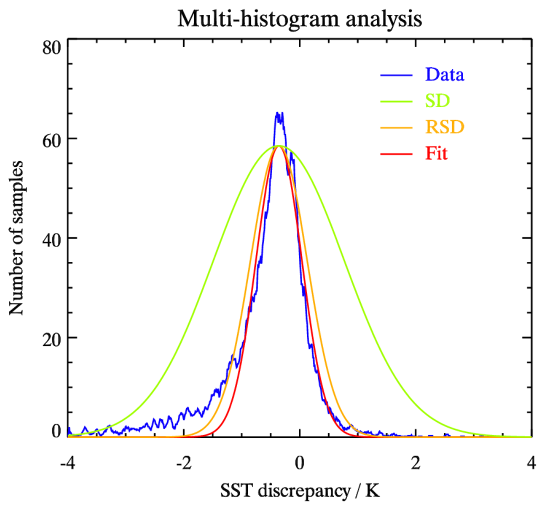


Figure 6-1: Distribution (in blue) of differences between the NOAA-19 AVHRR and drifting buoys for Australian waters. Over plotted are Gaussian PDFs from (in green) normal statistics, (in orange) robust statistics and (in red) a linear least square fit. The extreme outliers beyond the scale used in the plot force the width of the green distribution.

It is clear from Figure 6-1 that for our example distribution (with significant outliers) that the conventional statistics (green line) significantly overestimates the standard deviation of the distribution of the majority of data. The best representations for the central part of the distribution (within 1-sigma) are the linear fit (red line) and the robust statistics (orange line). From this discussion, it is clear that a distribution plot to ensure the chosen statistical model is appropriate for the data must accompany any statistical analysis.

6.2 Independence of validation activities

To ensure objective independent validation a high degree of independence in the valdaiton activity is required. This is true of both data and personnel.

6.3 Reference datasets

There are several potential reference data sources that can be used for assessing the quality of an SST CDR. Ship-borne radiometers are routinely calibrated and inter-calibrated and provide an estimate of the temperature at a defined depth (the skin temperature). Unfortunately, their sampling is regional and sparse, which sometimes leads to difficulties in relating a single radiometer observation to the SST value from the corresponding satellite pixel (the match-up). Although the small number of available radiometers mean that their coverage is limited compared to that offered by drifters, they are a vital source of reference data for CDR assessment as they provide the only reference data that are routinely traceable to national metrological standards. Our aim here is not to carry out an extensive review of available reference sources but to focus on their key aspects for SST CDR assessment.

The drifting buoy network provides greatly improved coverage compared to radiometers, except in the Southern Ocean and a few other areas such as upwelling zones. The coverage of the network is good because new devices are continually deployed but, unfortunately, they are not adequately calibrated prior to deployment and very few are recovered and recalibrated at the end of their deployment. Consequently, the overall accuracy of drifters is the subject of much debate and it is currently accepted to be ~ 0.2 K. Also, an obvious limitation is the fact that there is no knowledge or control over the depth at which the measurement is being made on account of the inevitable vertical motion of the buoys when floating in the surface waves.

The addition of Argo data to satellite SST validation is an important new development as although the floats measure SST_{depth} in a profile to no higher than ~3 m below the ocean surface, the intrinsic high accuracy of an Argo float (~ 0.05 K) means the total match-up uncertainty from comparing Argo SST_{depth} to satellite SST_{skin} is comparable to that of radiometers (which is a SST_{skin} to SST_{skin} comparison) at night and is considerably lower than both radiometers and drifters during the day. A new generation of ARGO floats is being tested with additional high frequency sampling in the upper 4 m of the ocean surface. The main issue with Argo is, like radiometers, their sampling, as each float only surfaces once every 10 days, and the uppermost temperature measurement has to be taken within the temporal window of the satellite overpass.

Other potential reference data moored buoys and conventional ship measurements from engine room intakes or hull-mounted sensors. The GTMBA array in the tropics are considered separately from other moored buoys because they are in the open ocean and far from the coastal regions which often present particular difficulties for the accurate measurements of SST from space, and where most other moored buoys are deployed. A summary of the primary reference datasets available for SST CDR assessment, and their estimated uncertainties, is given in Table 6-1.

Data type	Year	Coverage	SST*	Uncertainty
Ship-borne IR radiometers	1998 -	Repeated tracks in the Caribbean Sea, North Atlantic Ocean, North Pacific Ocean, and the Bay of Biscay; episodic deployments elsewhere in the world's oceans.	SSTskin	0.10 K
Argo floats	2000 -	Global[#] from ~ 2004 onwards.	SST-5m	0.05 K

GTMBA	1979 -	Tropical Pacific Ocean array completed in 1998; tropical Atlantic and Indian Ocean arrays installed later.	SST-1m	0.10 K
Drifting buoys	1991 -	Global# from ~ 2000 onwards.	SST-20cm	0.20 K

Table 6-1: Reference datasets availabe for SST CDR assessment and their estimated uncertainties

Data are not truly "global" but cover the majority of Earth's oceans. *Depths of SST are indicative, and often the actual depth for a single measurement is unknown.

In addition to the uncertainty of each potential dataset we also have to consider their temporal and spatial coverage, which has changed considerably over the years through both changes in the number and location of deployments and in sampling and reporting frequency. Figure 6-2 shows the temporal variation in the total number of available monthly reference data points for SST CDR assessment for the four reference datasets given in Table 6-1 as well as for Voluntary Observing Ships (VOS).

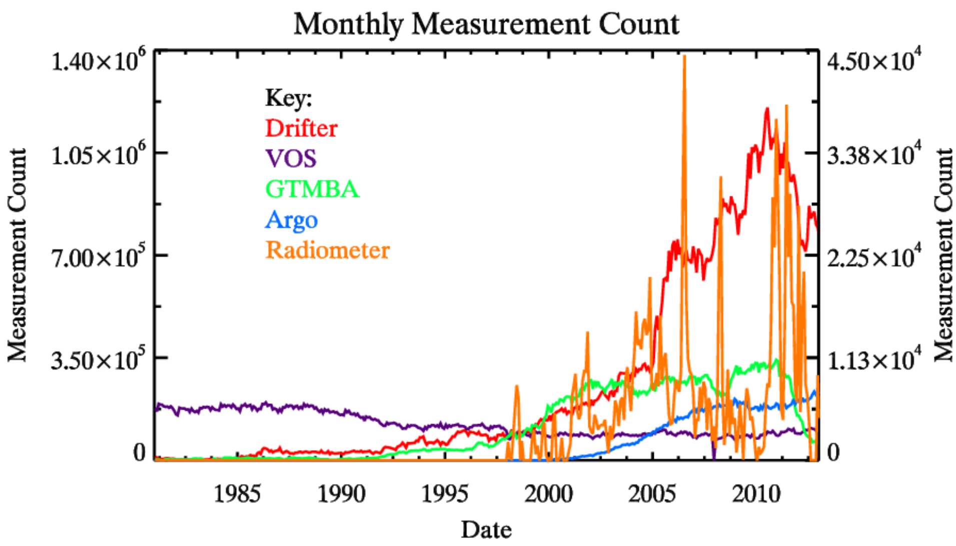


Figure 6-2: Temporal variation in total number of available monthly reference measurements from 1981 to 2013. Counts for drifters, VOS and GTMBA are shown on the primary y-axis and for Argo and radiometers on the secondary y-axis.

In Figure 6-2 the variation over time is considerable, with VOS measurements staying reasonably constant but deceasing slightly over time, whereas the number of drifting buoys has increased significantly since the year 2000 but has since declined. The increase in drifter measurements is a combination of both an increased reporting frequency and an increase in the number of deployments. Although small in number, but increasing in time, the high quality (see Table 6-1) measurements from Argo are an important reference dataset, particularly moving forward in time.

6.4 Match-up generation

The importance of specific quantitative measures relevant to the climate quality aspects of SST datasets cannot be over-emphasised, as these quantitative measures need to be comparable between the different sets of assessment information. In this section we mainly look at ways to assess how likely the SST at a given location may differ from the truth on average, i.e., representing the uncertainty associated with effects that are systematic. However, any such assessment has three important caveats that the user should be aware of:

1. As we have seen there are no globally distributed reference data with negligible errors against which satellite SST biases can be tested, and must use all available validation data;

2. Any averaging of SST must be done on appropriate space and time scales to avoid loss of the desired signal;

3. For comparability, strict independence of satellite and reference data is needed, which may restrict the comparisons that can be generated.

The Global Climate Observing System statement of requirements for satellite SST is that “accuracy” should attain “0.1 K over 100 km scales”, noting that “some … datasets may approach 0.1 K accuracy on a global average basis but have biases >0.5 K for many important regions”. This GCOS requirement therefore appears to be a statement about the acceptable systematic uncertainty as defined above. The relevant space-scale is 100 km (roughly 1° in latitude and longitude at the equator), according to GCOS. The present density of validation values available (mainly drifting buoys) is arguably insufficient to support assessment at this space scale, particularly prior to 2004. Merchant et al. have argued that satellite SST biases can be assessed using the current array of drifting buoys on space scales of 1000 km (roughly 1 degree at the equator), and the real limit is likely to be somewhere in-between and geographically variable.

Assessments on coarser space scales can be made using other validation data sets. Particularly important here are the uppermost SST measurements (typically at ~4 m depth) from Argo profiling floats, which provide acceptable global coverage from roughly 2004 onwards (see Figure 6.2). In addition, smaller regional scale, assessments can be carried out using the GTMBA and the long-term ship-borne radiometer deployments in the Caribbean Sea and the Bay of Biscay. The importance of these latter datasets should be noted as none of these data are currently used in the retrieval of satellite SSTs and are therefore remain independent.

6.5 Analysis methods

Once the generation of the matchup dataset is comeplte then several different analyses should be carried out, each one informing the valdation result.

6.5.1 Dependence plots

The objective of this analysis is to plot the dependence of the discrepancy between the reference dataset and the satellite as function of key aprameetrs that are known to have an impact on the SST retrieval. These include, latitude, wind pseed and total column water vapour.

6.5.2 Spatial variability

The objective of this analysis is to plot the dependence of the discrepancy between the reference dataset and the satellite as function of space (global and regional maps) and time (Hovmoller plots).

6.5.3 Histograms

The objective of this analysis is to plot the dependence of the discrepancy between the reference dataset as a histogram and peromr a linear least square fit of a normal Gaussian Proability Distribution Function (PDF) to the data.

6.5.4 Quantitative metrics

Once the mainly qualitative (visual) analysis is complete (for each reference dataset) several quantitative metrics can be computed. Table 3-1 shows example validation statistics from comparing ARC V1.0 AATSR data to various reference datasets. An adjustment has been made using a combined diurnal variability/skin model to account for differences in depth and time between the satellite and drifters, GTMBA and Argo; for radiometers only the time difference has been adjusted as they are estimates of the SST_{skin}.

The variation in number of match-ups between the various reference datasets is striking and also it is important to note that strictly speaking only the drifters and Argo results are directly comparable as these are “global” in nature. To correctly compare GTMBA and radiometer results, for example, one should extract “regional” drifter and Argo subsets corresponding to the spatial and temporal sampling of the GTMBA and radiometer match-ups (which is really only practical for drifters due to the low number of Argo match-ups). As well as reporting an overall systematic difference from drifting buoys, using the global median of the satellite minus drifting buoy SST difference, it must be put into context regarding any geophysical difference in the type of SST. For example, if the satellite SSTs are SST_{skin} with no skin/diurnal adjustment, then a skin-effect difference of order -0.17 K is to be expected in comparisons to drifting buoys. Also, only a single table covering all wind speeds is shown here, but results should be shown for both all wind speeds and for wind speeds > 6 ms^{-1} separately, as discussed above.

A further measure is then to examine the geographical variation in bias, which can be described by the standard deviation of median satellite minus drifting buoy SST differences at a specific space scale. As discussed above, limitations in the availability of reference data greatly influences what scale this can be done over. For example, the GHRSST CDAF currently recommends a scale of ~1000 km (averaging the data every 10 degree in latitude and longitude) for at least the period from 2005 onwards. At least for missions since around 2005, this measure should be assessable for most of the global oceans at this spatial scale.

One can then estimate the systematic difference to drifting buoys by first calculating the median ($\mu = \mathrm{median}(x_{satellite} - x_{in-situ})$) of each individual cell, and then calculating the standard deviation of μ across all cells as a measure of the systematic differences. To ensure that the results are reasonably sound statistically and to account for the uncertainty in the drifting buoys, the GHRSST CDAF recommends calculating σ_{cell} = $(\sigma_{ref})/\sqrt{n_{cell}}$ for each cell, and then rejecting cells where $2\sigma_{cell}$ > 0.1 K. This is equivalent to requiring n_{cell} > 16 and is expressed as such to indicate the rationale: namely that the uncertainty in the mean of the validation data should be smaller than the GCOS target with a high degree of confidence (~95%). It is also possible to use Argo measurements for determining the systematic difference for more recent years (2005 onwards). However, as the density of matches will be much less, the assessment has to be done on greater spatial scales.

As well as a systematic difference, the GHRSST CDAF also recommends calculating a non-systematic difference, i.e. the components of uncertainty that would remain after the systematic effects have been quantified. To estimate the non-systematic uncertainty using the same dataset(s) as for the systematic uncertainty, subtract the appropriate from each discrepancy: $d' = (x_{satellite} - x_{in-situ}) - \mu$ and then calculate the robust standard deviation of *d'*. This process will actually over-estimate the non-systematic uncertainty, because the measure includes effects arising from the imperfect *in situ* observations and true geophysical variability (differences from “point to pixel” comparisons, and difference in measurement times). The over-estimation is likely to be significant if the resulting value is close to about 0.2 K (for drifting buoys) or 0.1 K (for Argo).

6.6 Reporting of results

6.6.1 Journal articles

Whereever possible all valdaiton results should be published in the peer-vreviewed literature.

6.6.2 GHRSST User Guide and website

A summary of product quality is a key metric for data discovery and so all GHRSST data providers should provide a summary of their product valdaiton results to the GHRSST Project Office. This data will be provided ot the GHRSST archive, added to the GHRSST user manual and also published on the GHRSST website.

6.7 Traceability to SI

It is clear that the available reference data are not ideal for assessing the quality of long-term satellite-derived SST data records as they vary considerably in availability and quality. Temperature is one of the seven base SI units and establishing traceability to SI is an essential part of any assessment. A first attempt at providing a route to SI traceability was proposed by Minnett and Corlett and has subsequently evolved through a series of workshops held at the International Space Science Institute (ISSI) in Bern, Switzerland. A team of experts from the international SST community have proposed a method summarised in the flow diagram shown in Figure 6.3 as the main route to demonstrating SI traceability for long-term satellite-derived SST data records.

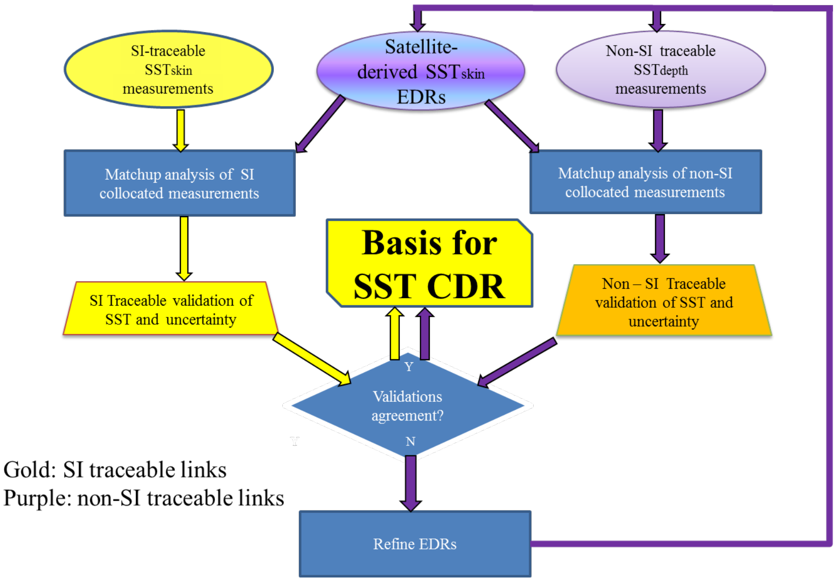


Figure 6-3: Flow diagram showing the route to demonstrating SI traceability for on orbit taken from ISSI meeting report. Ship-borne radiometer measurements, which are SI traceable, are included in the "match-up analysis of SI collocated measurements"

Figure 6-3 shows the key part of the process, which involves taking matchups between the satellite-derived SSTs and SI-traceable reference measurements, such as those from ship-borne IR radiometers, and comparing them to non-SI-traceable measurements such as those made by thermometers mounted on drifting buoys. The set of matchups with the ship radiometers provides traceability to SI-standards, and ideally this would be all that is required. The number of ship-borne radiometers is currently unable to provide adequate sampling of the parameter space that influences the satellite-SST retrieval and its uncertainties, and thus the much larger data set of matchups with subsurface measurements from buoys, although with higher uncertainties and with limited traceability, has also to be used.

The key decision process hinges on whether the uncertainty budgets derived from comparisons with SI-traceable and non-SI-traceable measurements are equivalent. If yes, it can be argued that the uncertainties derived from the much larger set of non-SI-traceable matchups can be used in the characterization of the long-term SST record. If not, it implies that the satellite-derived SSTs have uncertainties that are not well characterized in either sets of matchups, and the algorithms used in deriving the satellite SSTs should be improved. Metrics for the equivalence in the characteristics of the two uncertainty budgets have to be determined taking into account the properties of the two data sets and the methods used to generate the matchups.

Finally, when considering what is needed for reference data it is clear that it should be sufficient to 'sample' the complete measurement space and the critical factor is what the measurement space actually is. The measurement space should cover the dependence on key parameters and it should cover as much the entire global ocean as possible. Furthermore, it should be possible to sample the measurements space at regular intervals (ideally monthly) to identify any seasonal biases. This is challenging owing to the large variability in available reference data over time.

6.8 Long-term stability

The calculation of the long-term stability of satellite based SST data records is a mandatory step in the assessment of the data. For SST, we define stability is the degree of invariance over time of the mean error from systematic effects. GCOS requires a stability of "<0.03 K/decade over 100 km scales". To assess stability, we need a long-term reference dataset of known (and ideally) higher stability than that which is being assessed.

So far only one assessment of stability capable of being informative at the level of the GCOS requirement has been published. In Merchant et al.the stability of the long-term ARC record was assessed relative to components of the GTMBA over a 20-year period. Merchant et al. concluded that over the period 1994 to 2010 that regionally collocated ARC and GTMBA SSTs are stable, with better than 95% confidence, to within 0.005 K yr^{-1}. As ARC and GTMBA are two independent datasets it is reasonable to assume that the stability of 0.005 K yr^{-1} determined by Merchant et al. is an upper limit on the stability of the datasets individually for at least 1994 to 2010 in that specific region. As such, we have high confidence in the stability of the GTMBA for assessment of the SST_CCI datasets across this period in the specific region sampled by the GTMBA. We attribute the high stability of the GTMBA buoys to their routine maintenance, and, crucially, their pre- and post-deployment calibration to SI. However, as noted by Merchant et al. stability can only be directly assessed for equatorial latitudes using the GTMBA. Other regions do not yet have adequate in situ data characterisation in terms of uncertainty to establish trends although this is an on-going area for improvement and research.

Other options for stability assessment include the drifting buoy network but it is not known to be stable to the GCOS level (at least there are no published assessments) and is does not have spatial distributions that are stable in time. As such, drifting buoys cannot be used with confidence at the level of the GCOS stability requirement – although time-series of satellite-buoy differences should still be calculated in order to preclude major instability in the satellite record outside of tropical latitudes.

The Argo network of profiling floats has sensors stated to be of very high accuracy and stability, but as yet covers too brief a period to allow a rigorous assessment of decadal stability and is also subject to sparse sampling and regional distributions (as the floats are often entrained into ocean dynamical structures): the earliest the network is arguably complete (i.e. capable of providing global match-ups) is 2004 and techniques needed for exploiting the Argo network for stability assessment requires further study owing to (a) its relatively short lifetime and (b) its poor coverage over time at each location (so the data cannot be deseasonalised for example). Indeed, Argo is likely to provide the first global assessment of SST stability and consequently research into how to exploit the Argo data for stability is an urgent area to address.

The utility of ship-borne radiometers (1998 onwards) in areas of repeat ship tracks for stability assessment has not yet been established and is another area for on-going research. Radiometers may have a unique advantage above other reference datasets for stability assessment as uncertainties can be provided per measurement, thus avoiding the need for large numbers to reduce some of the uncertainty of the comparison between a point measurement and a satellite footprint.

The GTMBA moorings provide consistent SSTs in a defined geographical region across the whole time period of long-term satellite based SST datasets, with the number of observations available increasing over time due to (a) changes in reporting frequency (e.g. every hour to every minute) and (b) further deployments (e.g. addition of the Prediction and Research Moored Array in the Atlantic, PIRATA, and Research Moored Array for African-Asian-Australian Monsoon Analysis and Prediction, RAMA, arrays to the original Tropical Atmosphere Ocean, TAO, array). In Merchant et al. each GTMBA location used in their stability assessment had a minimum of 120 months with 5 or more match-ups between ARC and buoy over the period 1991–2009. In addition, only GTMBA locations that passed strict quality control procedures were used.

The data used by Merchant et al. was extracted from the International Comprehensive Ocean-Atmosphere Data Set (ICOADS) V2.5 dataset, which contains GTMBA data mainly obtained via the GTS system. The analysis is highly dependent on the long-term continuity of the GTMBA, particularly at sites with the longest available data records. Analysis that uses delayed mode GTMBA data is also possible and one would expect a larger range of sites to be used compared to Merchant et al. as the improved reporting frequency will reduce the uncertainty on individual locations.

The use of the GTMBA for stability assessment is now an essential part of activities to analyse long-term satellite derived SST records. As such, the GHRSST CDAF has such an analysis included as a core activity. The methodology for stability assessment defined in the GHRSST CDAF is simplified relative to the approach of (Merchant et al. [8]) and has been trialled within the ESA SST_CCI project. The idea of the CDAF is to use a simplified – but scientifically robust – method that can easily be applied across multiple datasets to provide comparable metrics.

Unlike those applied by in Merchant et al., step-detection techniques are not specified in the GHRSST CDAF, as these require significant resources and expertise to implement. A consequence, of not using step-detection techniques is that step changes have to be identified visually/subjectively, with a corresponding chance of both steps being missed and steps being falsely imputed. Development of step-detection methods, ideally applicable across CDR domains to identify correlations between geophysical variables, is very much needed.

The approaches presented so far for stability assessment rely on (a) estimating the stability of the reference dataset and (b) use the rule of large numbers to reduce the uncertainty due to random effects. If we had a reference dataset with known uncertainties per measurement, then we would not have to use the rule of large match-ups for assessment and instead could use individual measurements and their uncertainties and assess their degree of equivalence as shown schematically in Figure 6.4. In Figure 6.4 we show three hypothetical difference datasets (1) in black, the satellite data, (2) in red, a poorly characterised reference dataset whereby a mean uncertainty for the entire dataset has to be used for each measurement, and (3) in green, a well characterised reference dataset in which each measurement has its own uncertainty. For our idealised case one can see that any stability assessment carried out between the black and red points would lead one to conclude the satellite time series has a very low stability, whereas a stability assessment carried out between the black and green points would lead one to conclude the satellite time series has a very good stability when the measurement uncertainties are considered.

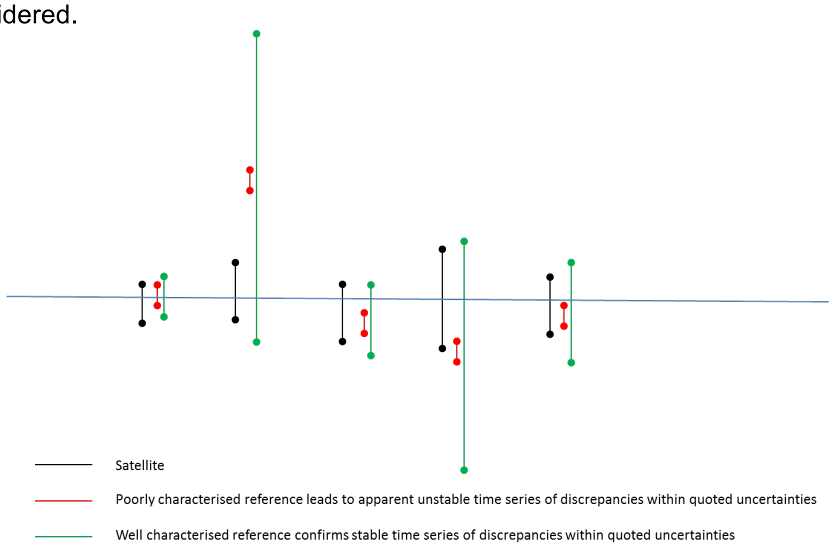


Figure 6-4: Schematic example of three time series of match-ups between satellite and reference data shown as uncertainties (the dispersion of the values that could reasonably be attributed to the measurand). The satellite data is shown in black and two different instances of the reference data are shown in red, for a poorly characterised reference dataset whereby a mean uncertainty is used for each measurement, and in green, for a well-characterised reference dataset in which each measurement has its own uncertainty.

So far none of the available reference datasets can provide individual measurement uncertainties. However, efforts are well underway to quantify them for ship-borne radiometers.

7 Validation of uncertainties

Given the significant variations in available reference data (in space, time and accuracy) the only pragmatic approach to providing uncertainties for a satellite SST CDR is to generate the uncertainties using an uncertainty model, which itself needs to be validated. A key step in evaluating CDRs of SST is therefore to assess both the SST and its quoted uncertainty. The principal approach to validation of uncertainties is to examine the distribution of satellite-reference SST differences as a function of uncertainty. This method has been trialled by Lean and Saunders in their assessment of the ARC dataset.

In an ideal case, the standard deviation of the differences between the satellite SST and a reference SST would equal the satellite uncertainty, i.e.

$$\sigma_{sat-ref} = \sigma_{sat}$$

However, the reference data has its own uncertainties to consider, as discussed in Section 2.3. Consequently the standard deviation of the differences between the satellite SST and a reference SST is really a combination of both the uncertainty in the satellite SST and the uncertainty in the reference SST, i.e.

$$\sigma_{sat-ref} = \sqrt{\sigma_{sat}^2 + \sigma_{ref}^2}$$

As we discussed earlier, there are of course the other terms to consider relating to:

- The difference in spatial sampling (a point reference measurement versus a satellite pixel);
- The difference in depth of the measurements;
- The difference in time of the measurements.

Such an approach naturally considers the uncertainty due to environmental effects related to the homogeneity of a region/process. For example, validation in a region dominated by strong SST fronts at low wind speed will mean the first term (spatial sampling) will be systematic for any one single match-up.. However, as the number of match-ups increases the uncertainty will reduce by $1/\sqrt{N}$ as you sample the variability at multiple locations (unless you always sample on one side of a front, say). Consequently, the effect is considered to be a pseudo-random term across a set of validation data and not systematic. Likewise, in an area of strong solar radiation and low wind speed the second term (difference in depth) would be systematic for any one match-up. Therefore, these latter terms can be reduced to being << 0.1 K in the mean through the use of a depth/time adjustment, large number of match-ups (to reduce pseudo-random terms) and through like versus like (SST_{skin} versus SST_{skin} or SST_{depth} versus SST_{depth}) comparisons.

An idealised Uncertainty Validation plot, assuming validation against data with Gaussian errors with a standard deviation of 0.2 K, is shown in Figure 7.1. Vertical lines span -1 to +1 standard deviation of discrepancy, for data binned into 0.1 K bins of estimated satellite SST uncertainty. When the satellite SST uncertainty is small, the SD of discrepancy is dominated by the in situ uncertainty. For large satellite SST uncertainty, the SD of discrepancy approaches the estimated uncertainty of the satellite. The Dotted line gives the locus of the results if the satellite SST uncertainty is perfectly estimated. Deviations from the dotted line indicate biases in uncertainty estimation

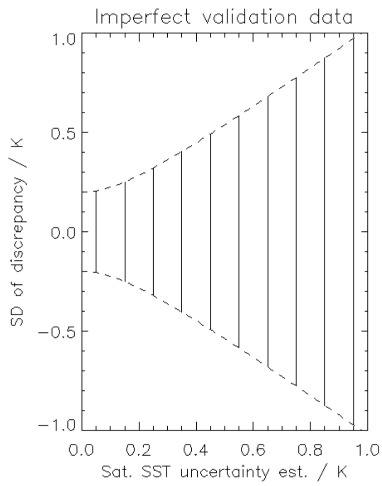


Figure 7-1: Idealised Uncertainty Validation plot, assuming validation against data with Gaussian errors with a standard deviation of 0.2 K. Vertical lines span -1 to +1 standard deviation of discrepancy, for data binned into 0.1 K bins of estimated satellite SST uncertainty. When the satellite SST uncertainty is small, the SD of discrepancy is dominated by the in situ uncertainty. For large satellite SST uncertainty, the SD of discrepancy approaches the estimated uncertainty of the satellite. The Dotted line gives the locus of the results if the satellite SST uncertainty is perfectly estimated. Deviations from the dotted line indicate biases in uncertainty estimation.

One can easily see in Figure 7.1 that at low satellite uncertainties the standard deviation of the differences is dominated by the uncertainty in the reference data and as satellite uncertainties grow the satellite uncertainty dominates the statistics, as the reference uncertainty becomes a less significant contribution to the total uncertainty. In fact, the uncertainty of the reference data can dominate the statistics at low satellite uncertainties meaning it may not be possible to validate the uncertainty model once this limit has been reached. Also, it is clear that as uncertainties are added in quadrature, the geophysical terms assumed to be small will be more significant at lower satellite uncertainties and a “geophysical limit” will be present even for reference data with uncertainties << 0.1 K.

7.1 Uncertainty verification

The provision and validation of product uncertainties derived from a physical-based uncertainty model is deemed essential for the assessment of any long-term SST record owing to the variations in reference data over time; arguably it is essential for any long-term geophysical measurement record. However, we still have to find a way to provide some sort of confidence in the measurand and its associated uncertainties even when no reference data is available. This is the idea behind the ‘functional’ validation whereby we use knowledge gained from comparisons to referenced data to
Within the ESA SST_CCI project an attempt was made to provide verification maps to indicate where independent verification of product uncertainties could provide confidence that they were of the right order of magnitude. The verification maps were generated by computing the % difference between the calculated and theoretical RSD of the differences between the various ESA SST_CCI datasets and drifting buoys. The comparisons were carried out across the full range of uncertainties at each 15 degrees of latitude and longitude (taking into account uncertainties in the drifting buoy data). The median % difference for each latitude/longitude cell is then scaled to give an indication of verification according to:

- Very high – uncertainties are confirmed to be within 20% of their quoted values
- High – uncertainties are confirmed to be within 20% - 40% of their quoted values
- Medium – uncertainties are confirmed to be within 40% - 60% of their quoted values
- Low – uncertainties are confirmed to be within 60% - 80% of their quoted values
- Very low – uncertainties are confirmed to be within 80% - 100% of their quoted values

A “not verifiable” category was also included where it was not possible to independently verify the product uncertainties using the reference dataset. It is important to recognise that it did not mean that the product uncertainties should not be used in these cases, just that they could not be confirmed independently.

An example uncertainty verification map for the ESA SST_CCI analysis dataset is shown in Figure 7-2. The coverage is very good with very few unverified regions. On average the uncertainties are of high quality compared to the reference dataset and in general regions of medium and low quality occur in areas that contain low numbers of drifting buoys.

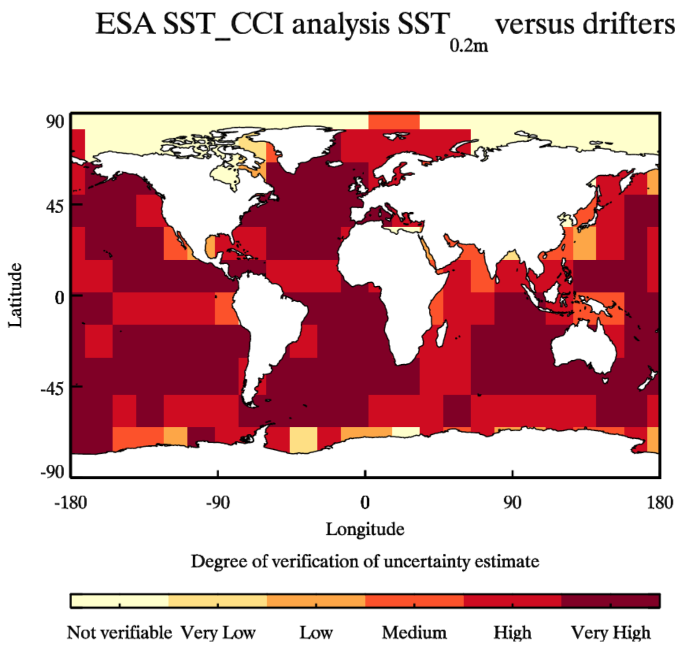


Figure 7-2: Verification maps for ESA SST_CCI SST_{depth} uncertainties assessed using drifting buoy SST_{depth}. This plot shows the degree to which the SST CCI product uncertainties can be verified using independent reference data. It should not be taken as an indication of SST CCI product data quality and is intended to help the user interpret their own results from applying product uncertainties in their analysis.

The initial attempt at uncertainty verification from the ESA SST_CCI project has its limitations in that the distribution of sea ice has not been factored in so verification results in Polar Regions are going to be lower than they actually are. Also, the maps do not distinguish between cases where (a) the degree of verification is low due to a low number of match-ups from cases where (b) the degree of the verification is low due to the measured uncertainties being over-estimated or under-estimated.

Nevertheless, these maps can provide users with a unique assessment of the product uncertainty quality and will allow them to scale the product uncertainties should they wish to do so in regions of low degree of verification. A method for implementing “functional” verification can then be added using match-ups with similar dependence on Total Column Water Vapour (TCWV), Solar Zenith Angle (SZA) or Aerosol Optical Depth (AOD) etc. that can be identified using dummy locations (i.e. no reference data). It is reasonable to assume the uncertainty model is correlated between such locations and therefore the uncertainties are of the right order of magnitude even though this cannot be independently verified.

8 Summary, conclusions and recommendations

The validation of satellite SSTs is a challenge in many dimensions, with multiple solutions that vary in space, time and quality, and on the application of use. In this document, we have presented a review of the problems encountered when carrying out satellite SST validation and have presented an agreed protocol for how to do the validation. Significant variations in the quality and coverage of available reference data mean approaches that use uncertainty models, which are validated, are recommended. Assessment methods should “follow the physics” (e.g., should compare like with like) and a validation uncertainty budget should be defined covering all known effects.

Determining the stability of the sensor is a fundamental requirement and the development of practical step-detection methods that are applicable to a range of geophysical variables is a recommended research priority. New methods to establish SI traceability are needed, especially for non-radiometric reference data, and a better understanding of the uncertainties of the various reference data sources is required.

9 Annexe A: Example validation analysis

To be added once main text agreed with GHRSST Science Team.

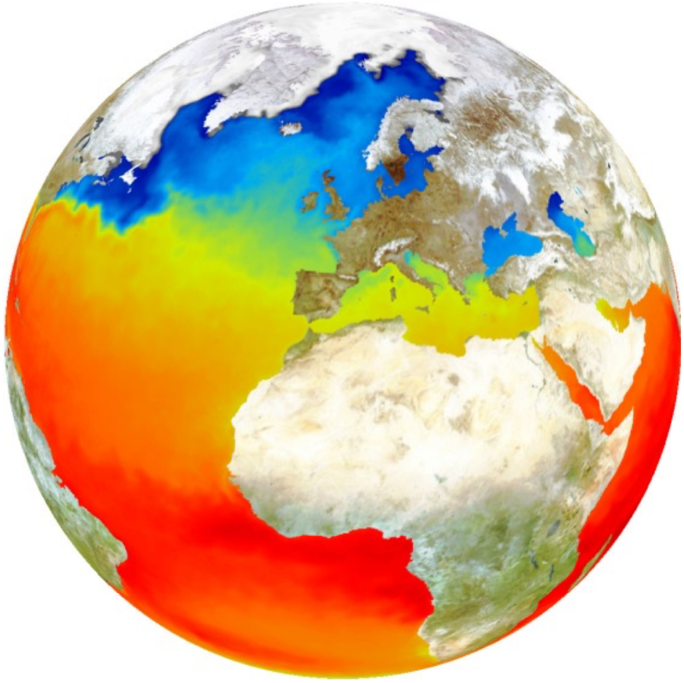
How to find out more about GHRSST:

A complete description of GHRSST together with all project documentation can be found at the following web spaces:

Main GHRSST portal	https://www.ghrsst.org
GHRSST GDAC (rolling archive)	http://ghrsst.jpl.nasa.gov
GHRSST LTSRF (Archive)	http://ghrsst.nodc.noaa.gov
GHRSST HRDDS (diagnostics)	http://www.hrdds.net
GHRSST MDB (validation)	http://www.ifremer.fr/matchupdb
GHRSST GMPE (L4 ensembles)	

http://ghrsst-pp.metoffice.com/pages/latest_analysis/sst_monitor/daily/ens/index.html

GHRSST data discovery	http://ghrsst.jpl.nasa.gov/data_search.html
GHRSST data visualisation (EU)	http://www.naiad.fr
GHRSST data visualisation (USA)	http://podaac-tools.jpl.nasa.gov/dataminer/



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